

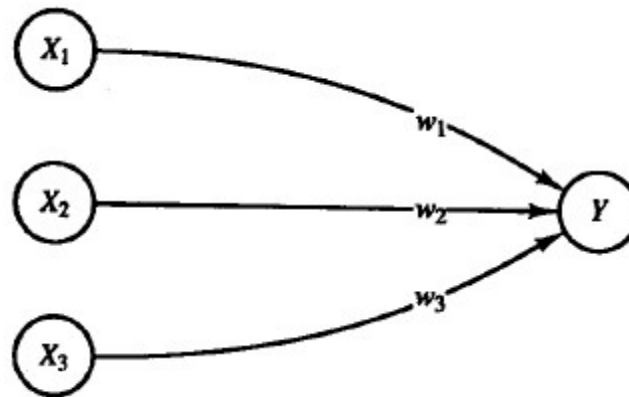
What is a NN?

- ❑ The development of artificial neural networks began approximately 60 years ago, motivated by a desire to try both to understand the brain and to emulate some of its strengths.
- ❑ Artificial neural networks have been developed as generalizations of mathematical models of human cognition or neural biology, based on the assumptions that:
 - ❑ 1- Information processing occurs at many simple elements called *neurons*.

What is a NN?

- ❑ 2- Signals are passed between neurons over connection *links*.
- ❑ 3- Each connection link has an associated *weight*, which, in a typical neural net, multiplies the signal transmitted.
- ❑ 4- Each neuron applies an *activation function* (usually nonlinear) to its net input (sum of weighted input signals) to determine its output signal.

A simple Neuron



$$y_in = w_1x_1 + w_2x_2 + w_3x_3.$$

$$y = f(y_in)$$

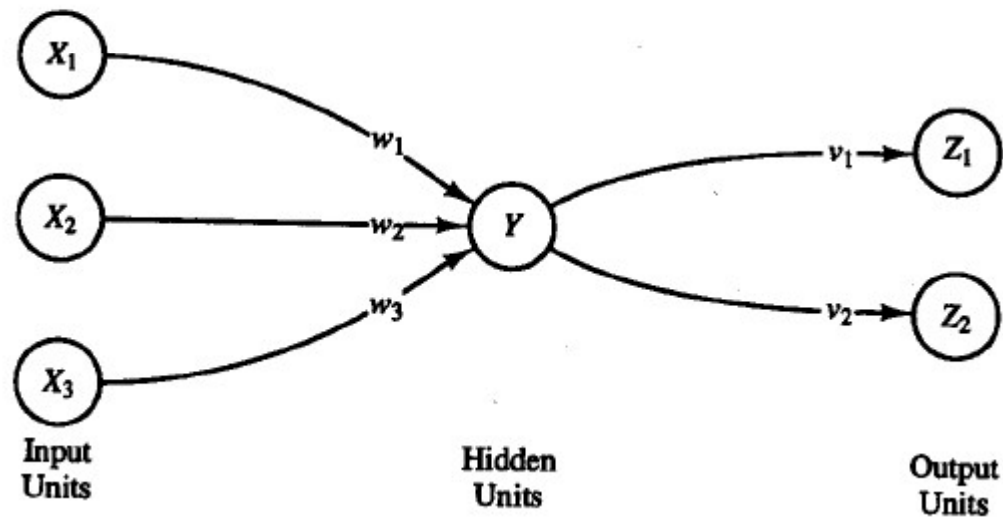
$$f(x) = \frac{1}{1 + \exp(-x)},$$

What is a NN?

- ❑ A neural network is characterized by:
 - its pattern of connections between the neurons (called its architecture),
 - its method of determining the weights on the connections (called its training, or learning, algorithm), and
 - its activation function.

Very simple NN

- Now suppose further that neuron Y is connected to neurons Z1 and Z2 , with weights v_1 and v_2 , respectively.



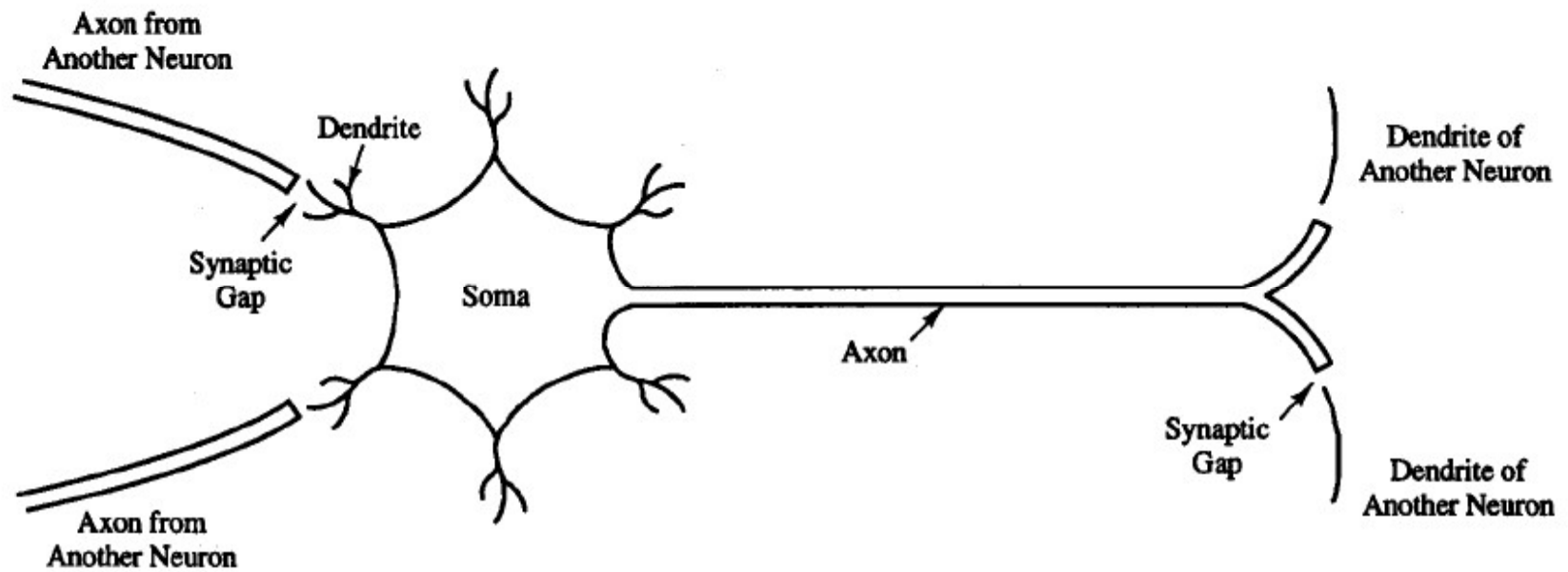
Very simple NN

- ❑ Neuron Y sends its signal y to each of these units. However, in general, the values received by neurons Z1 and Z2 will be different, because each signal is scaled by the appropriate weight, v_1 or v_2 .
- ❑ In a typical net, the activations z_1 , and z_2 of neurons Z1 , and Z2 depend on inputs from several or even many neurons, not just one.

Biological NN

- ❑ There is a close analogy between the structure of a biological neuron (i.e., a brain or nerve cell) and the processing element (or artificial neuron) presented in the rest of this book.
- ❑ A biological neuron has three types of components that are of particular interest in understanding an artificial neuron: its *dendrites*, *soma*, and *axon*.

Biological NN



Biological NN

- ❑ The many dendrites receive signals from other neurons.
- ❑ The signals are electric impulses that are transmitted across a synaptic gap by means of a chemical process.
- ❑ The action of the chemical transmitter modifies the incoming signal (typically, by scaling the frequency of the signals that are received) in a manner similar to the action of the weights in an artificial neural network.

Biological NN

- ❑ The soma, or cell body sums the incoming signals. When sufficient input is received, the cell *fires*; that is, it transmits a signal over its axon to other cells.
- ❑ It is often supposed that a cell either fires or doesn't at any instant of time, so that transmitted signals can be treated as *binary*.

Biological NN

- ❑ Several key features of the processing elements of artificial neural networks are suggested by the properties of biological neurons:
 - 1) The processing element receives many signals.
 - 2) Signals may be modified by a weight at the receiving synapse.
 - 3) The processing element sums the weighted inputs.
 - 4) Under appropriate circumstances (sufficient input), the neuron transmits a single output.
-

Biological NN

- 5) The output from a particular neuron may go to many other neurons (the axon branches).
- 6) Information processing is local.
- 7) Memory is distributed:
 - a. Long-term memory resides in the neurons' synapses or weights.
 - b. Short-term memory corresponds to the signals sent by the neurons.

Biological NN

- 8) A synapse's strength may be modified by experience.
- 9) Neurotransmitters for synapses may be excitatory or inhibitory.

Biological NN

- ❑ Biological neural systems are fault tolerant in two respects:
 - a) we are able to recognize many input signals that are somewhat different from any signal we have seen before. An example of this is our ability to recognize a person in a picture we have not seen before or to recognize a person after a long period of time.

Biological NN

- b) we are able to tolerate damage to the neural system itself. Humans are born with as many as 100 billion neurons. Most of these are in the brain, and most are not replaced when they die.
- In spite of our continuous loss of neurons, we continue to learn.

NN Applications

- ☐ Signal Processing.
 - Noise cancelation in telephone system.
 - ☐ Control
 - truck backer-upper
 - ☐ Pattern recognition
 - Handwritten characters
 - ☐ Medicine
 - Instance physician
 - ☐ Speech production and recognition
 - NETtalk
-

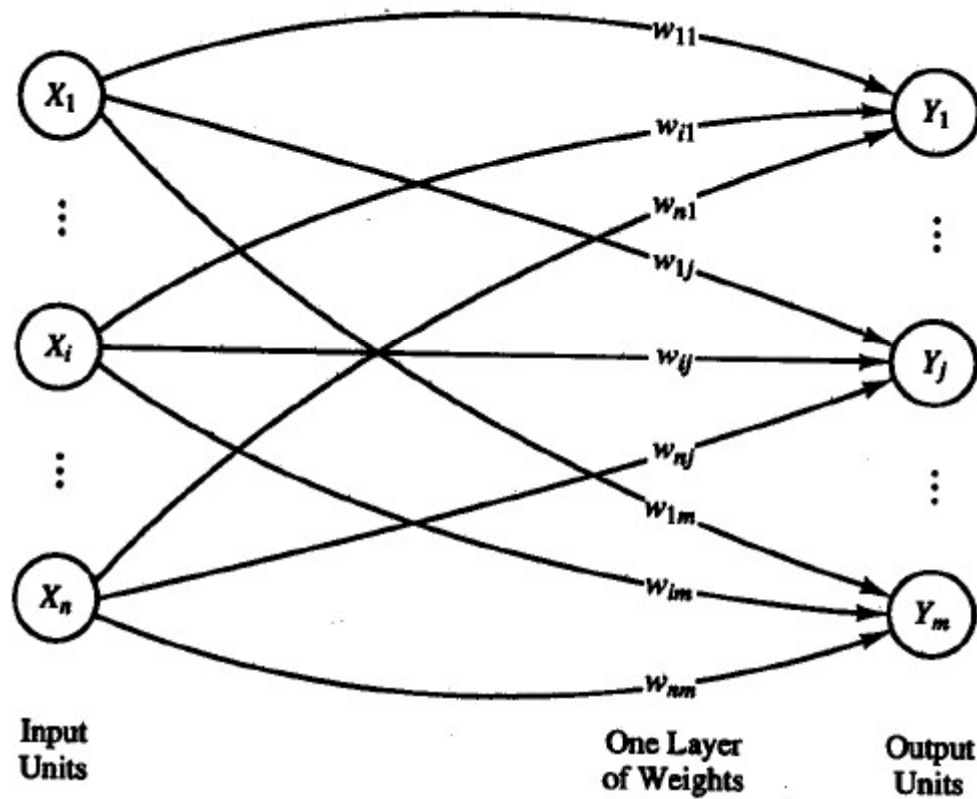
Typical Architectures

- ❑ The arrangement of neurons into layers and the connection patterns within and between layers is called the net architecture.
- ❑ Typically, neurons in the same layer behave in the same manner.
- ❑ To be more specific, in many neural networks, the neurons within a layer are either fully interconnected or not interconnected at all.
- ❑ Neural nets are often classified as single layer or multilayer.

Single layer Net

- ❑ A single-layer net has one layer of connection weights.
- ❑ the units can be distinguished as **input units**, which receive signals from the outside world, and **output units**, from which the response of the net can be read.
- ❑ For **pattern classification**, each output unit corresponds to a particular category to which an input vector may or may not belong.
- ❑ For **pattern association**, the same architecture can be used, but now the overall pattern of output signals gives the response pattern associated with the input signal that caused it to be produced.

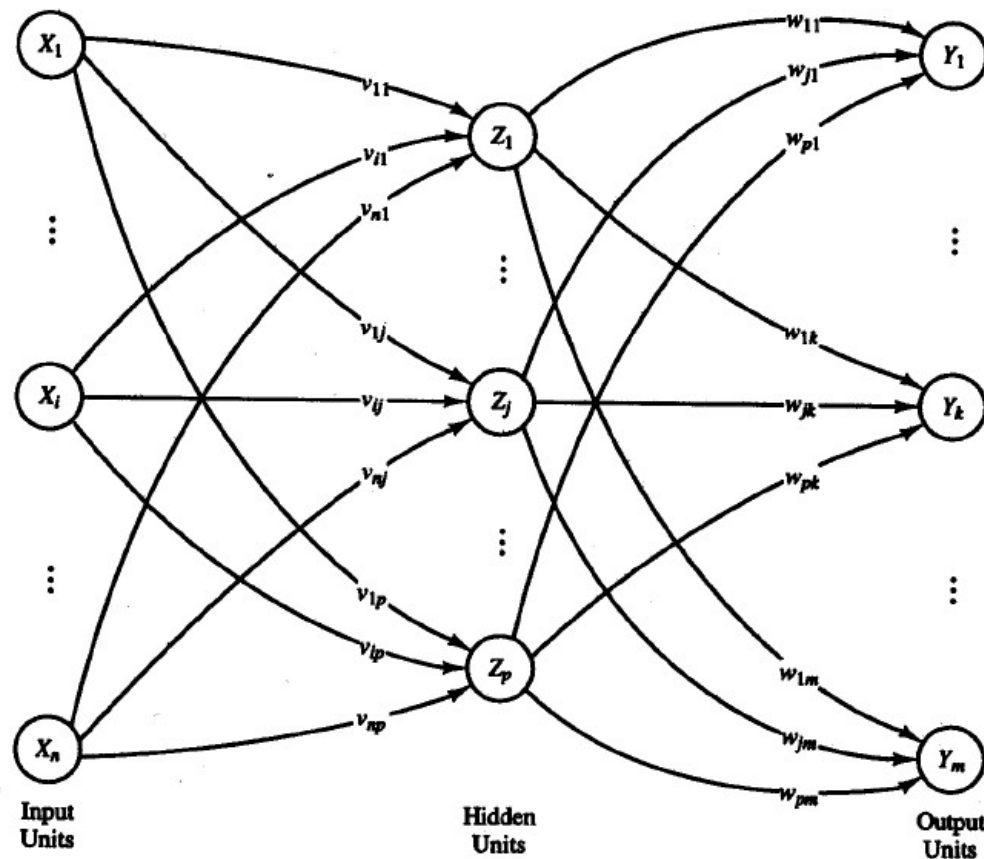
A single-layer net



Multi-layer Net

- ❑ the same type of net can be used for different problems, depending on the interpretation of the response of the net.
- ❑ On the other hand, more complicated mapping problems may require a multilayer network.
- ❑ A multilayer net is a net with one or more layers (or levels) of nodes (the so called hidden units) between the input units and the output units.
- ❑ Multilayer nets can solve more complicated problems than can single-layer nets, but training may be more difficult.

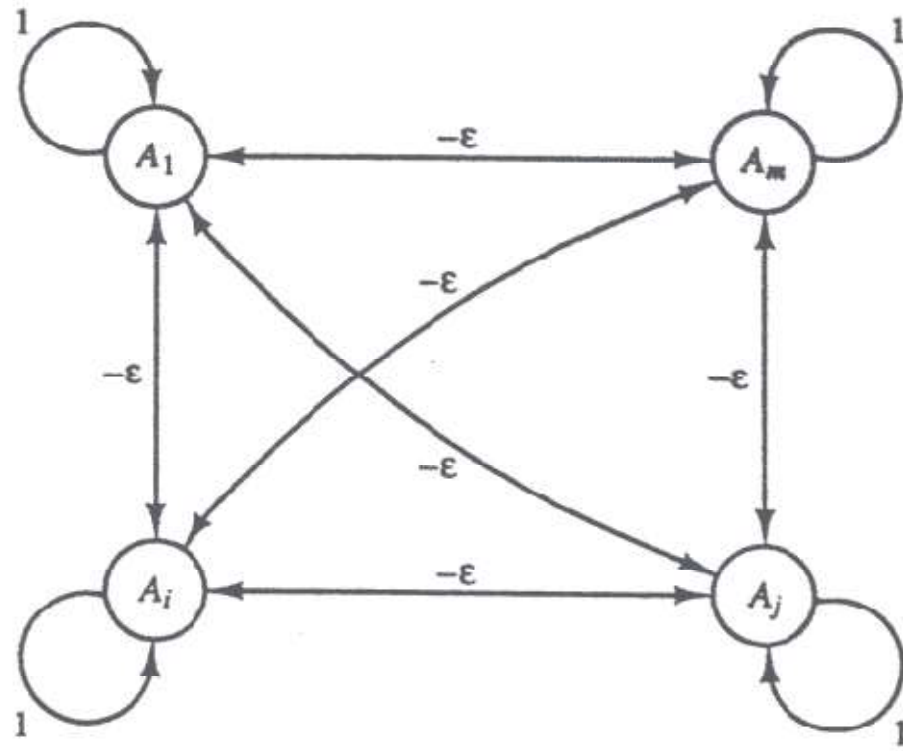
A multilayer net



Competitive layer

- ❑ A **competitive layer** forms a part of a large number of neural networks.
- ❑ Typically, the interconnections between neurons in the competitive layer are not shown in the architecture diagrams for such nets.
- ❑ The **winner-take-all** competition, and **MAXNET** are based on competition.

Competitive layer



Setting the Weights

- ❑ in addition to the architecture, the method of setting the values of the weights (**training**) is an important distinguishing characteristic of different neural nets
- ❑ For a neural network three types of training is used
 - Supervised
 - unsupervised
 - Fixed

Supervised training

- ❑ In perhaps the most typical neural net setting, training is accomplished by presenting a sequence of training vectors, or patterns, each with an associated target output vector.
- ❑ The weights are then adjusted according to a learning algorithm.
- ❑ This process is known as supervised training.
- ❑ the output is a bivalent element, say, either 1 (if the input vector belongs to the category) or -1 (if it does not belong).

Supervised training

- ❑ **Pattern association** is another special form of a mapping problem, one in which the desired output is not just a "yes" or "no," but rather a pattern.
- ❑ A neural net that is trained to associate a set of input vectors with a corresponding set of output vectors is called an **associative memory**.
- ❑ If the desired output vector is the same as the input vector, the net is an **autoassociative** memory; if the output target vector is different from the input vector, the net is a **heteroassociative** memory.

Unsupervised training

- ❑ **Self-organizing** neural nets group similar input vectors together without the use of training data to specify what a typical member of each group looks like or to which group each vector belongs.
- ❑ A sequence of input vectors is provided, but no target vectors are specified.
- ❑ The neural net will produce an **exemplar** (representative) vector for each cluster formed.

Fixed-weight nets

- ❑ Still other types of neural nets can solve **constrained optimization problems**.
- ❑ Such nets may work well for problems that can cause difficulty for traditional techniques, such as problems with conflicting constraints (i.e., not all constraints can be satisfied simultaneously).
- ❑ The **Boltzmann machine** (without learning) and the continuous Hopfield net can be used for constrained optimization problems.

Activation Functions

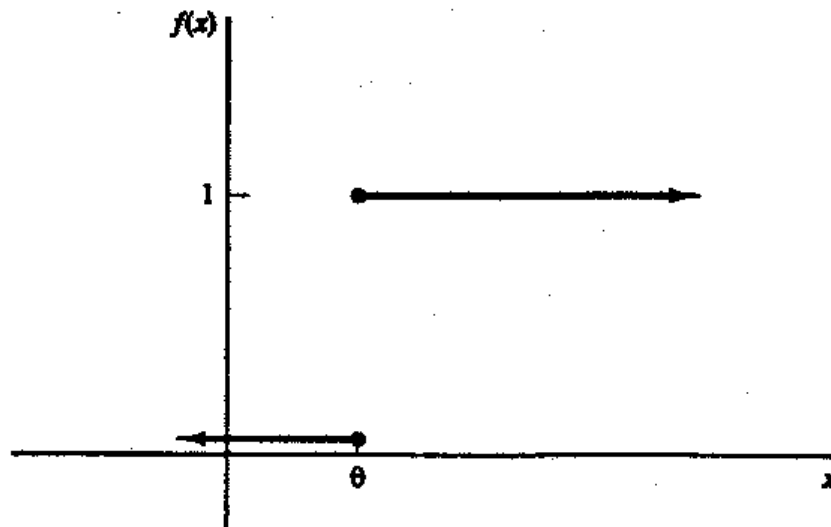
- ❑ the basic operation of an artificial neuron involves summing its weighted input signal and applying an activation function.
- ❑ Typically, the same activation function is used for all neurons in any particular layer of a neural net, although, this is not required.
- ❑ In most cases, a nonlinear activation function is used.

Activation Functions

- ☐ Binary step function.
- ☐ Binary sigmoid.
- ☐ Bipolar sigmoid.

Binary step function

- ❑ Single-layer nets often use a step function to convert the net input, which is a continuously valued variable, to an output unit that is a binary (1 or 0) or bipolar (1 or - 1) signal.

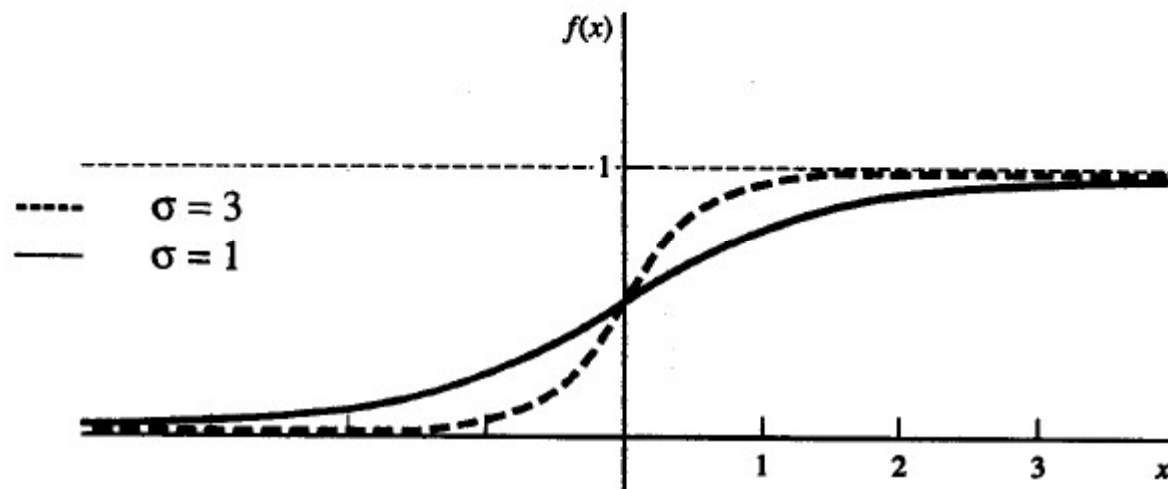


$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

Binary sigmoid

- ❑ Sigmoid functions (S-shaped curves) are useful activation functions.
- ❑ The logistic function and the hyperbolic tangent functions are the most common.
- ❑ They are especially advantageous for use in neural nets trained by backpropagation, because the simple relationship between the value of the function at a point and the value of the derivative at that point reduces the computational burden during training.

Binary sigmoid

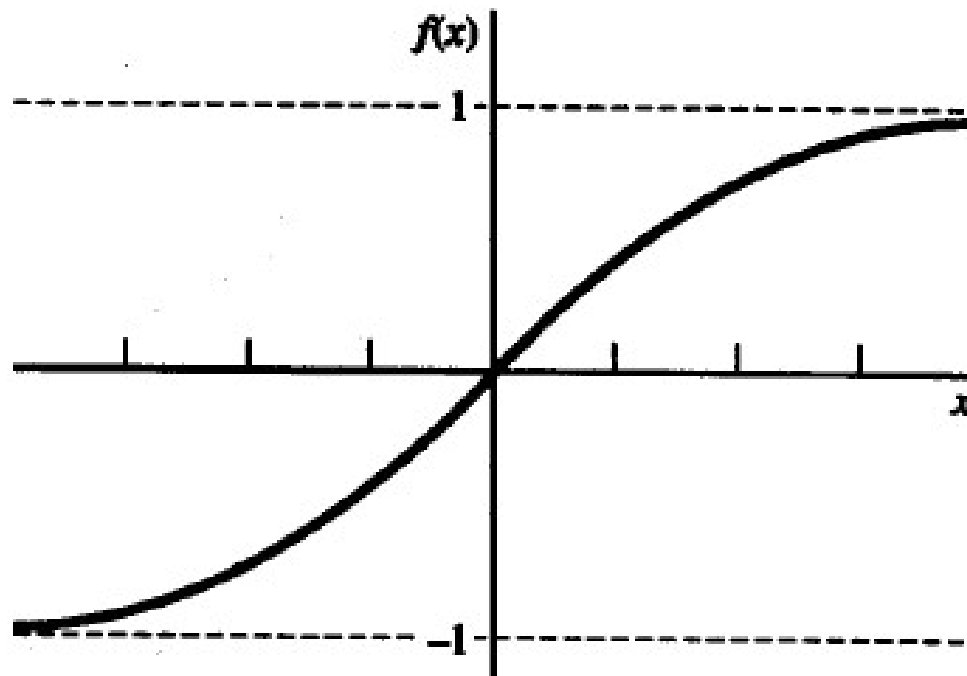


Bipolar sigmoid

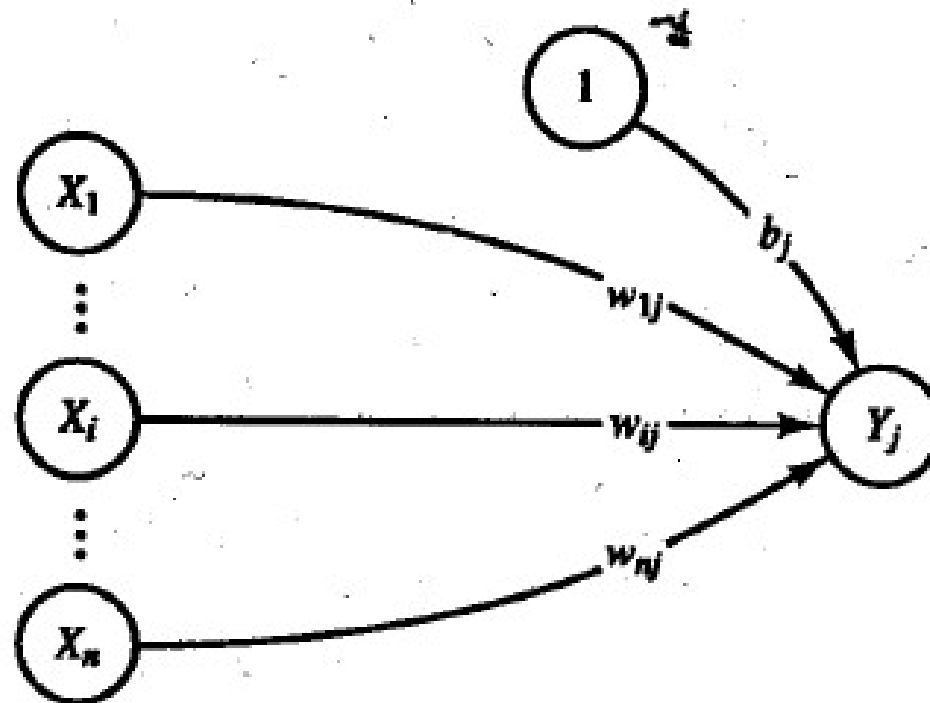
- ❑ The logistic sigmoid function can be scaled to have any **range of values that is appropriate** for a given problem.
- ❑ The most common range is from - 1 to 1; we call this sigmoid the bipolar sigmoid.

$$\begin{aligned}g(x) &= 2f(x) - 1 = \frac{2}{1 + \exp(-\sigma x)} - 1 \\&= \frac{1 - \exp(-\sigma x)}{1 + \exp(-\sigma x)} \\g'(x) &= \frac{\sigma}{2} [1 + g(x)][1 - g(x)].\end{aligned}$$

Bipolar sigmoid



Summary of Notation



Summary of Notation

x_i, y_j Activations of units X_i, Y_j , respectively:

For input units X_i ,

$x_i = \text{input signal}$;

for other units Y_j ,

$y_j = f(y_{in_j})$.

w_{ij} Weight on connection from unit X_i to unit Y_j :

Beware: Some authors use the opposite convention, with w_{ij} denoting the weight from unit Y_j to unit X_i .

b_j Bias on unit Y_j :

A bias acts like a weight on a connection from a unit with a constant activation of 1 (see Figure 1.11).

y_{in_j} Net input to unit Y_j :

$$y_{in_j} = b_j + \sum_i x_i w_{ij}$$

Summary of Notation

W Weight matrix:

$$W = \{w_{ij}\}.$$

$w_{.j}$ Vector of weights:

$$w_{.j} = (w_{1j}, w_{2j}, \dots, w_{nj})^T.$$

This is the j th column of the weight matrix.

$\| \mathbf{x} \|$ Norm or magnitude of vector $\mathbf{x}$.

θ_j Threshold for activation of neuron Y_j :

A step activation function sets the activation of a neuron to 1 whenever its net input is greater than the specified threshold value θ_j ; otherwise its activation is 0 (see Figure 1.8).

$\mathbf{s}$ Training input vector:

$$\mathbf{s} = (s_1, \dots, s_l, \dots, s_n).$$

$\mathbf{t}$ Training (or target) output vector:

$$\mathbf{t} = (t_1, \dots, t_j, \dots, t_m).$$

Summary of Notation

x Input vector (for the net to classify or respond to):

$$\mathbf{x} = (x_1, \dots, x_i, \dots, x_n).$$

Δw_{ij} Change in w_{ij} :

$$\Delta w_{ij} = [w_{ij} \text{ (new)} - w_{ij} \text{ (old)}].$$

α Learning rate:

The learning rate is used to control the amount of weight adjustment at each step of training.

Matrix manipulation

$$y_{in_j} = \mathbf{x} \cdot \mathbf{w}_j$$

$$= \sum_{i=0}^n x_i w_{ij}$$

$$= w_{0j} + \sum_{i=1}^n x_i w_{ij}$$

$$= b_j + \sum_{i=1}^n x_i w_{ij}$$

McCULLOCH- PITTS

- ❑ It is one of the first of neural nets.
- ❑ McCulloch-Pitts neurons may be summarized as:
 - The activation of a McCulloch-Pitts neuron is **binary**. That is, at any time step, the neuron either **fires** (has an activation of 1) or **does not fire** (has an activation of 0).
 - neurons are connected by directed, weighted paths.
 - A connection path is **excitatory** if the weight on the path is positive; otherwise it is **inhibitory**. All excitatory connections into a particular neuron have the same weights.

McCULLOCH- PITTS

- All excitatory connections into a particular neuron have the same weights.
- Each neuron has a **fixed threshold** such that if the net input to the neuron is greater than the threshold, the neuron fires.
- The threshold is set so that **inhibition is absolute**. That is, any nonzero inhibitory input will prevent the neuron from firing.
- It takes one time step for a signal to pass over one connection link.

McCULLOCH- PITTS

The threshold for unit Y is 4

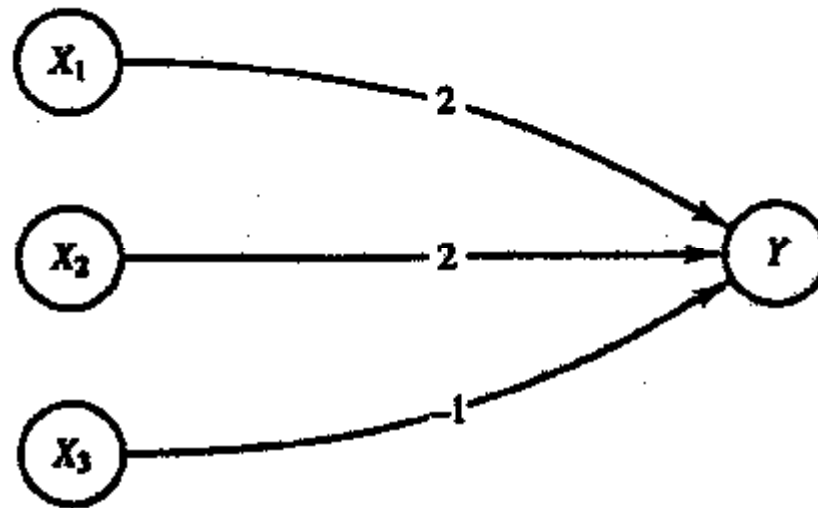


Figure 1.12 A simple McCulloch-Pitts neuron Y.

McCULLOCH- PITTS

In general form:

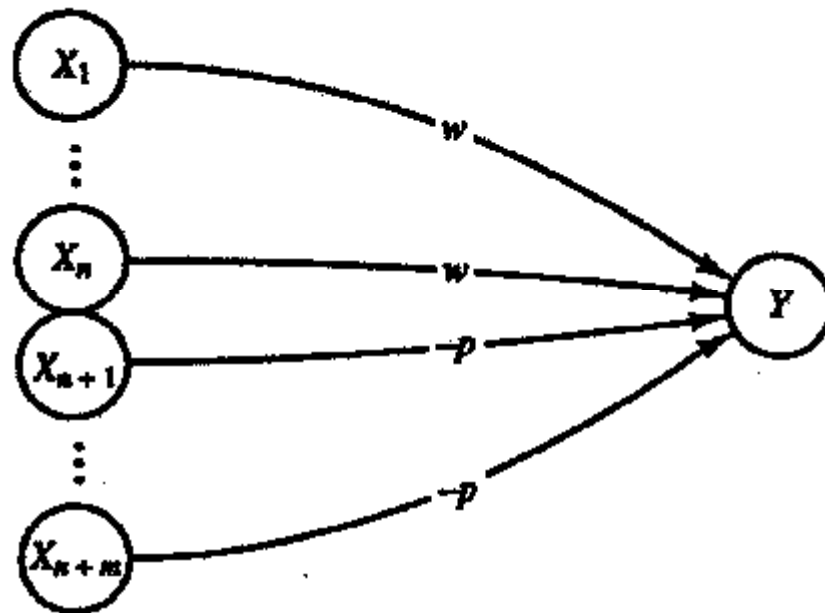


Figure 1.13 Architecture of a McCulloch-Pitts neuron Y .

McCULLOCH- PITTS

- The condition that inhibition is absolute requires that for the activation function satisfy the inequality:

$$\theta > nw - p.$$

- Y will fire if it receives k or more excitatory inputs and no inhibitory inputs, where:

$$kw \geq \theta > (k - 1)w.$$

Applications

- ❑ The condition that inhibition is absolute requires that for the activation function satisfy the inequality:
- ❑ The weights for a McCulloch-Pitts neuron are set, together with the threshold for the neuron's activation function, so that the neuron will perform a simple logic function.
- ❑ Using these simple neurons as building blocks, we can model any function or phenomenon that can be represented as a logic function.

AND

x_1	x_2	$\rightarrow$	y
1	1		1
1	0		0
0	1		0
0	0		0

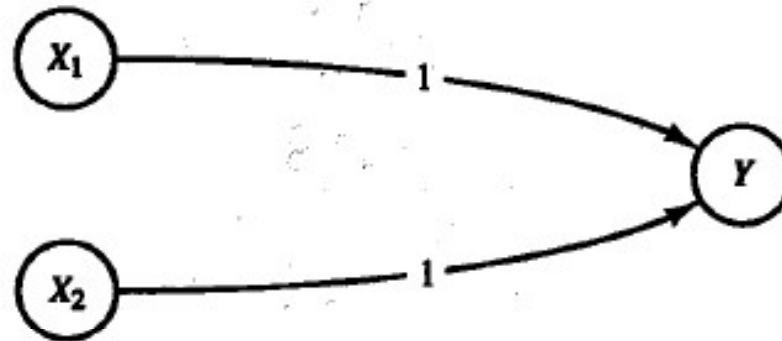


Figure 1.14 A McCulloch-Pitts neuron to perform the logical AND function.

OR

x_1	x_2	$\rightarrow$	y
1	1		1
1	0		1
0	1		1
0	0		0

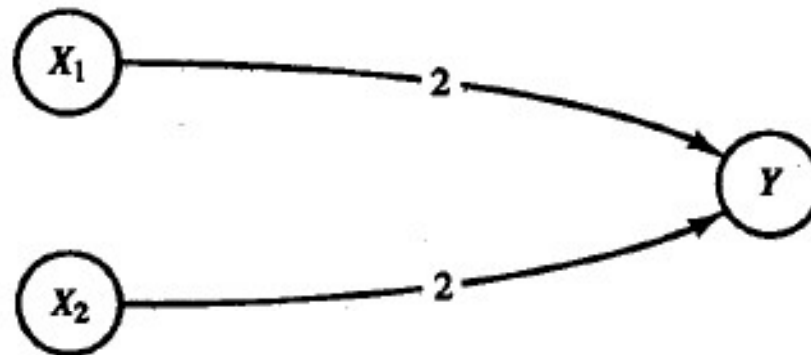
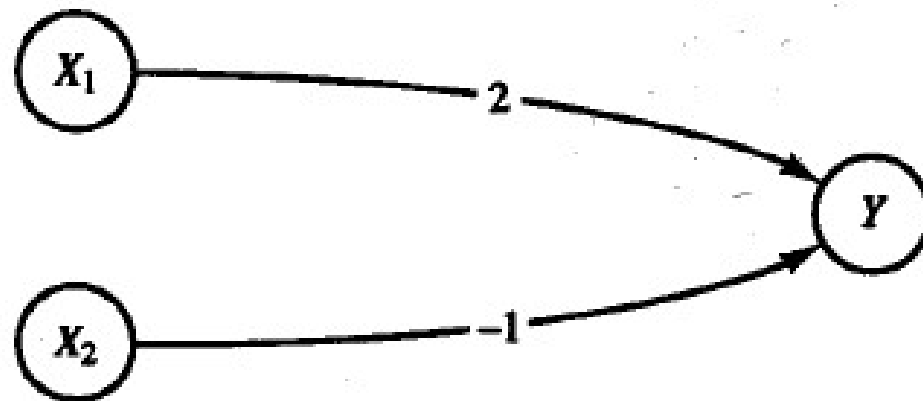


Figure 1.15 A McCulloch-Pitts neuron to perform the logical OR function.

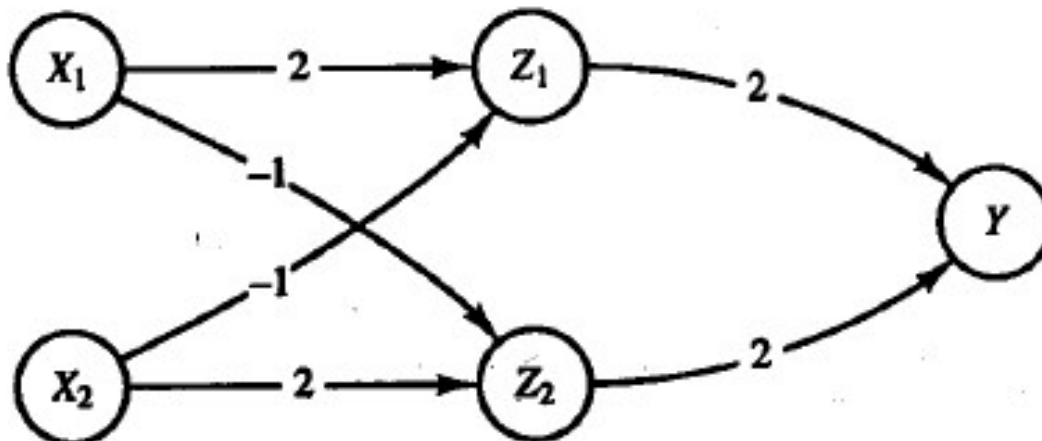
AND NOT

x_1	x_2	$\rightarrow$	y
1	1		0
1	0		1
0	1		0
0	0		0



XOR

x_1	x_2	$\rightarrow$	y	$x_1 \text{ XOR } x_2 \leftrightarrow (x_1 \text{ AND NOT } x_2) \text{ OR } (x_2 \text{ AND NOT } x_1).$
1	1		0	$z_1 = x_1 \text{ AND NOT } x_2$
1	0		1	$z_2 = x_2 \text{ AND NOT } x_1.$
0	1		1	
0	0		0	$y = z_1 \text{ OR } z_2.$



Hot & Cold

- ❑ if a cold stimulus is applied to a person's skin for a very short period of time, the person will perceive heat.
- ❑ However, if the same stimulus is applied for a longer period, the person will perceive cold.
- ❑ In the figure, neurons X1 and X2 represent receptors for heat and cold, respectively, and neurons Y1 and Y2 are the counterpart preceptors.
- ❑ Neurons Z1 and Z2 are auxiliary units needed for the problem
- ❑ Each neuron has a threshold of 2.

Hot & Cold

- ❑ The desired response of the system is that cold is perceived if a cold stimulus is applied for two time steps, i.e.,

$$y_2(t) = x_2(t - 2) \text{ AND } x_2(t - 1).$$

- ❑ heat be perceived if either a hot stimulus is applied or a cold stimulus is applied briefly (for one time step) and then removed.

$$y_1(t) = \{x_1(t - 1)\} \text{ OR } \{x_2(t - 3) \text{ AND NOT } x_2(t - 2)\}.$$

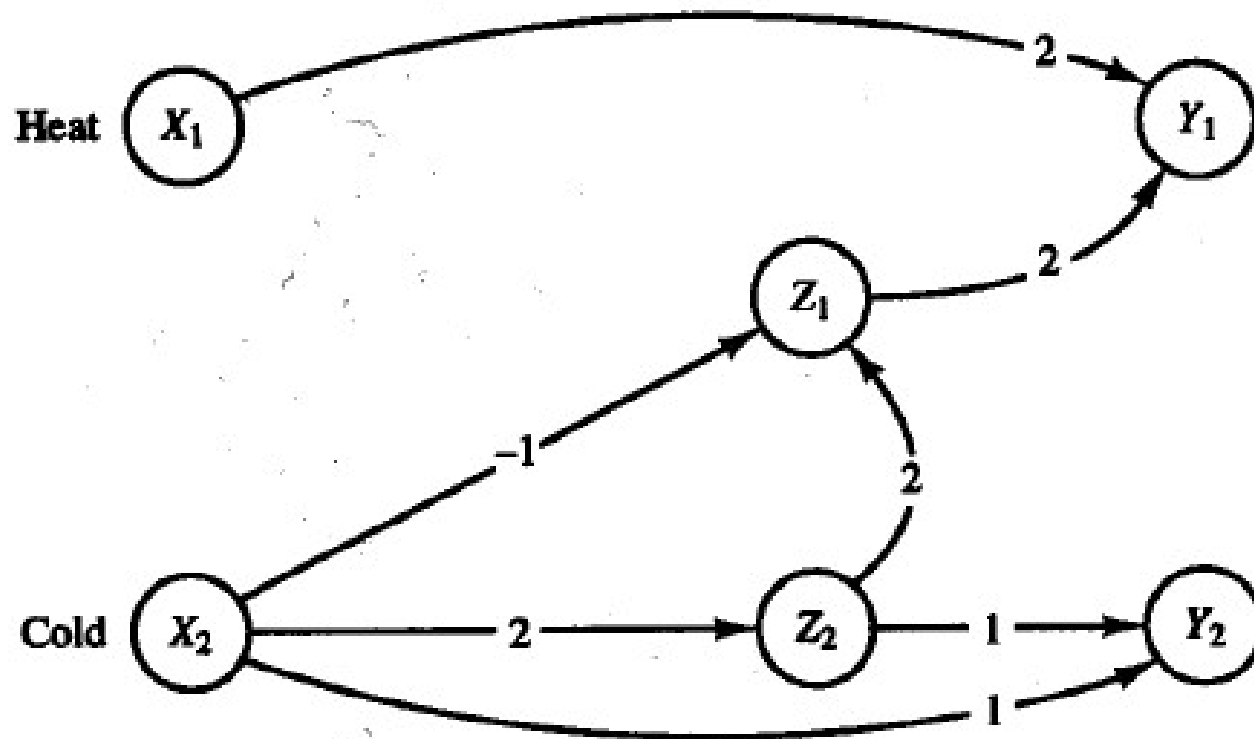
Hot & Cold

$$z_2(t - 2) = x_2(t - 3).$$

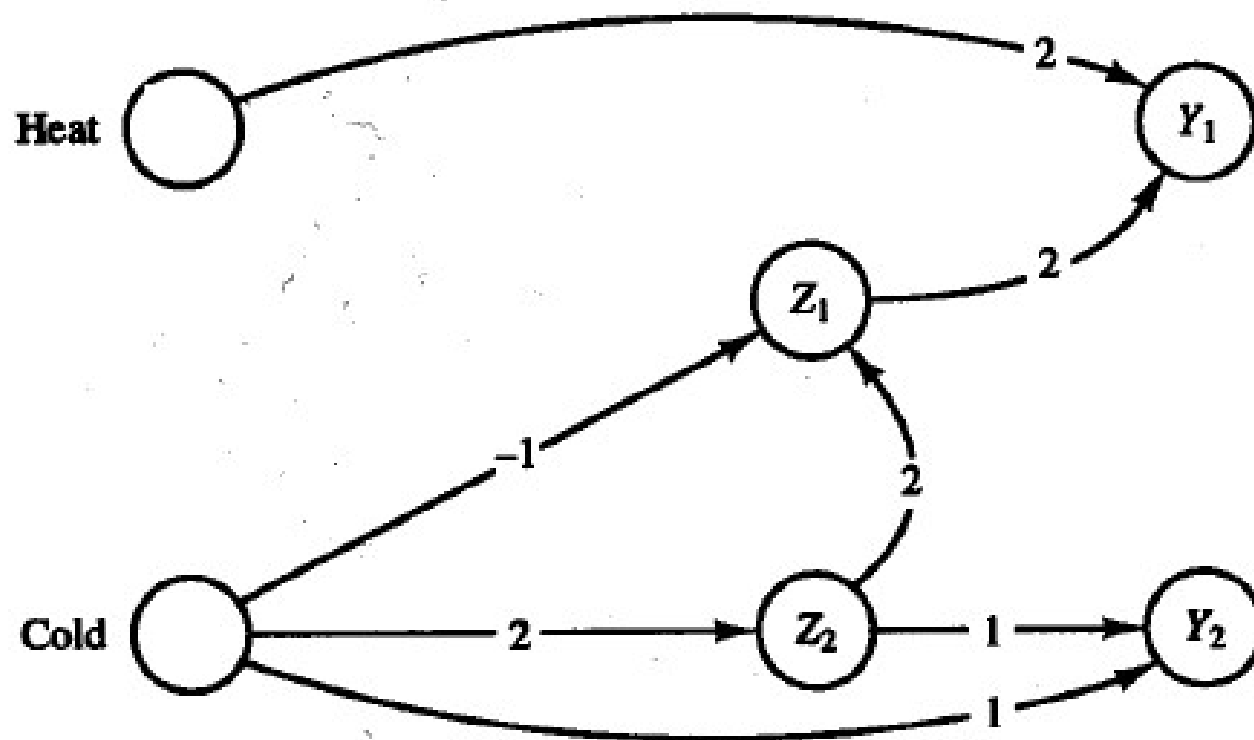
$$z_1(t - 1) = z_2(t - 2) \text{ AND NOT } x_2(t - 2).$$

$$y_1(t) = x_1(t - 1) \text{ OR } z_1(t - 1).$$

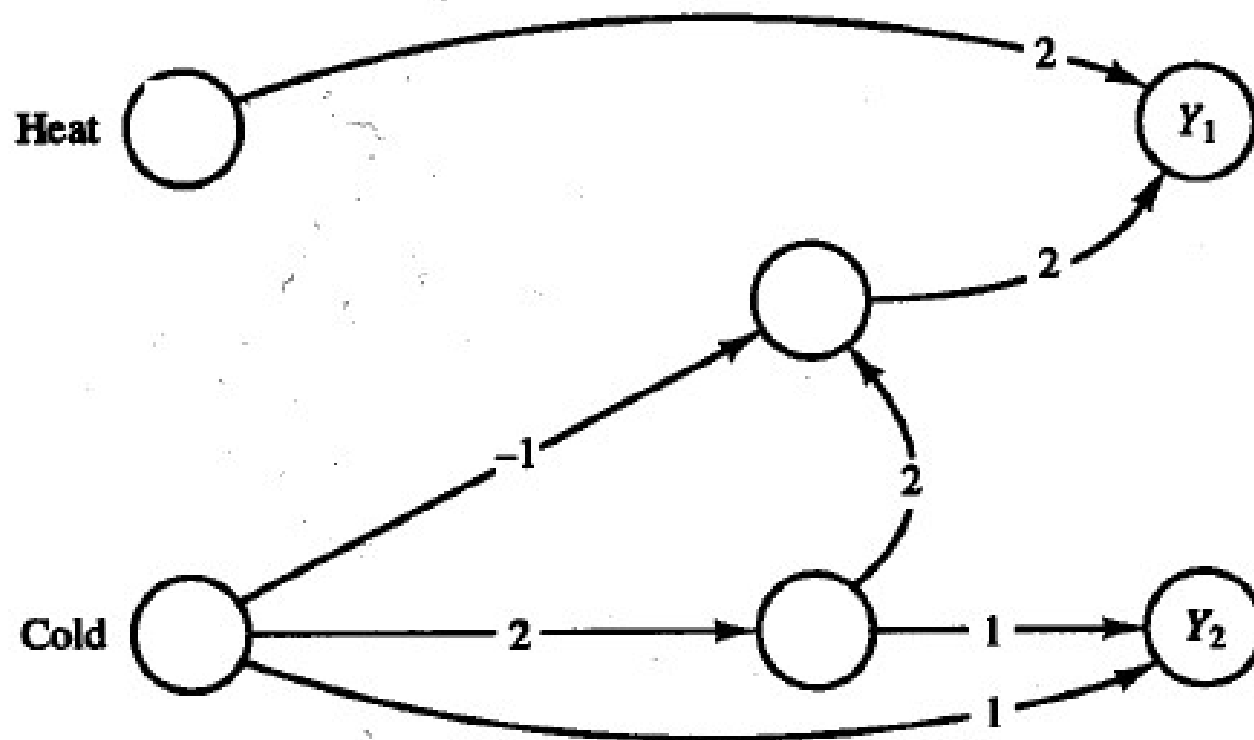
Hot & Cold



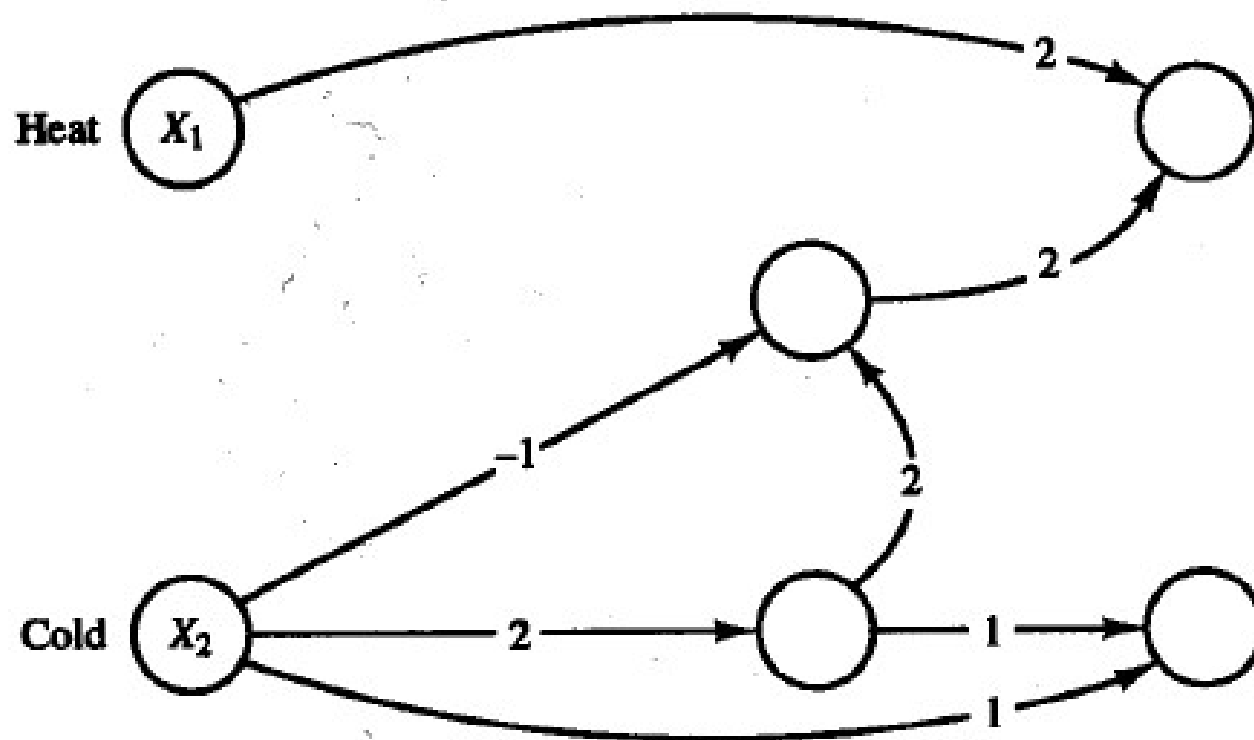
Cold Stimulus (one step)



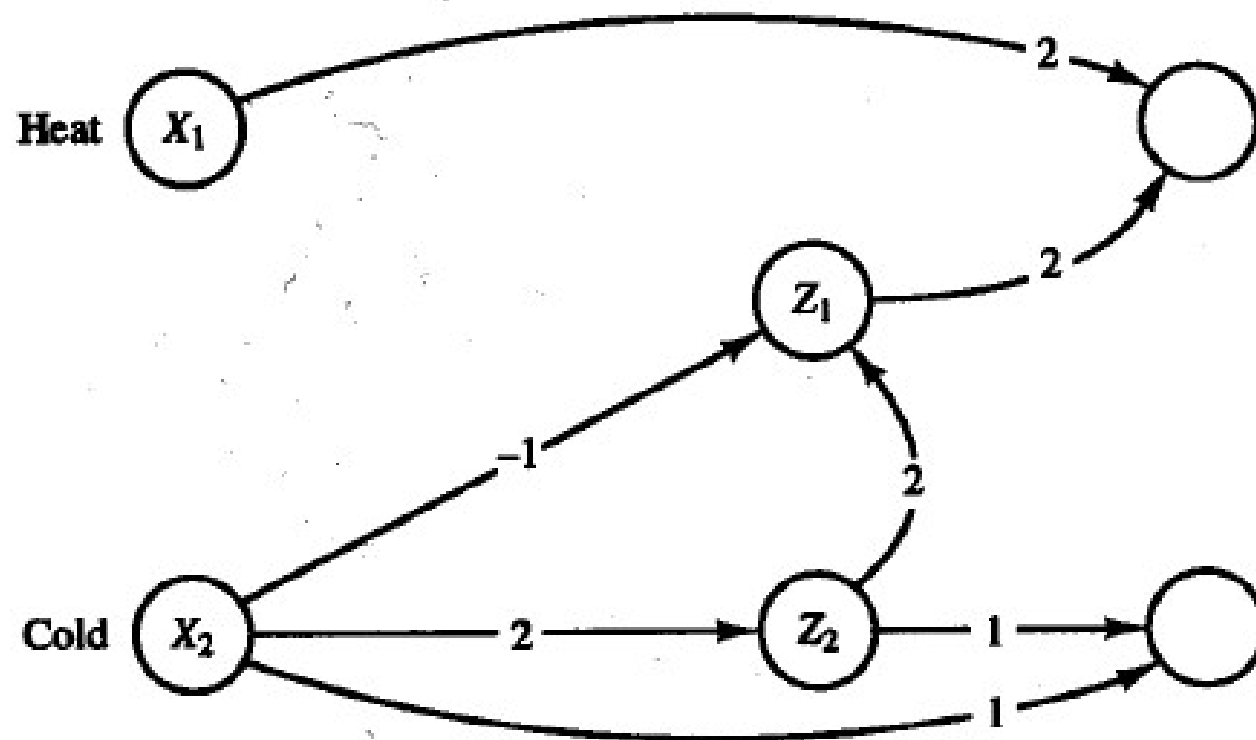
$t=1$



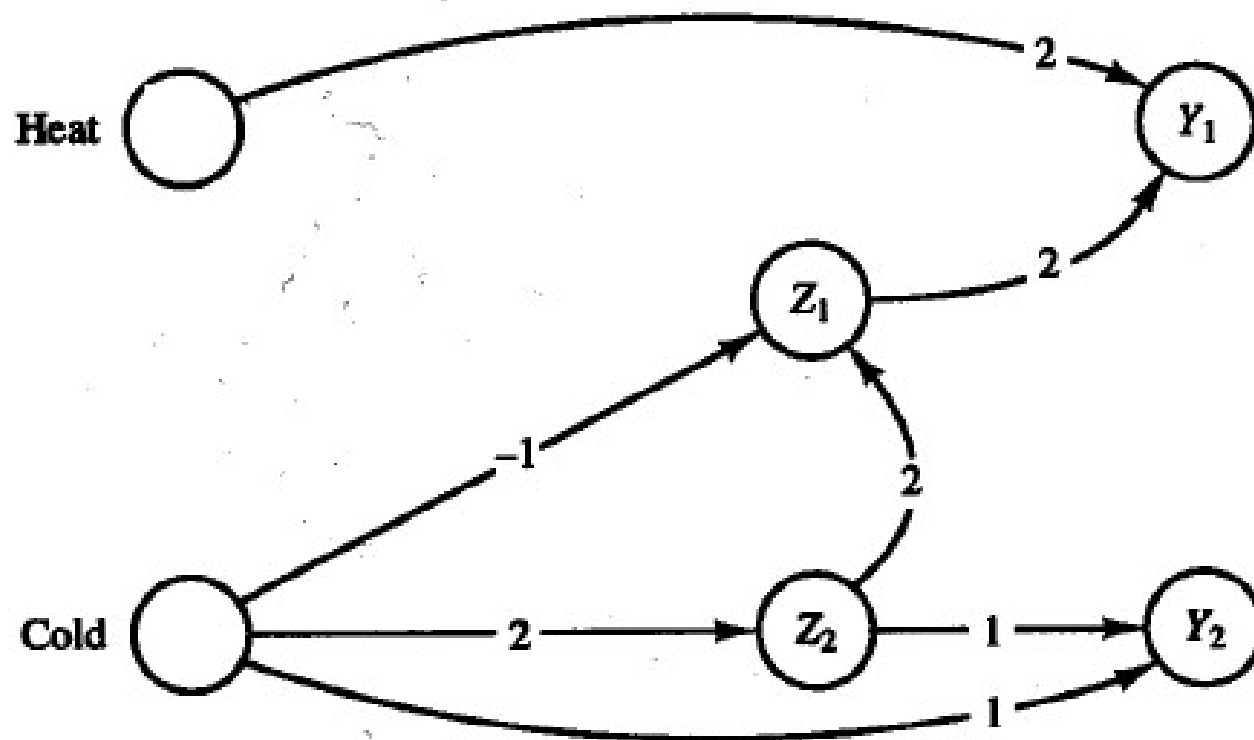
$t=2$



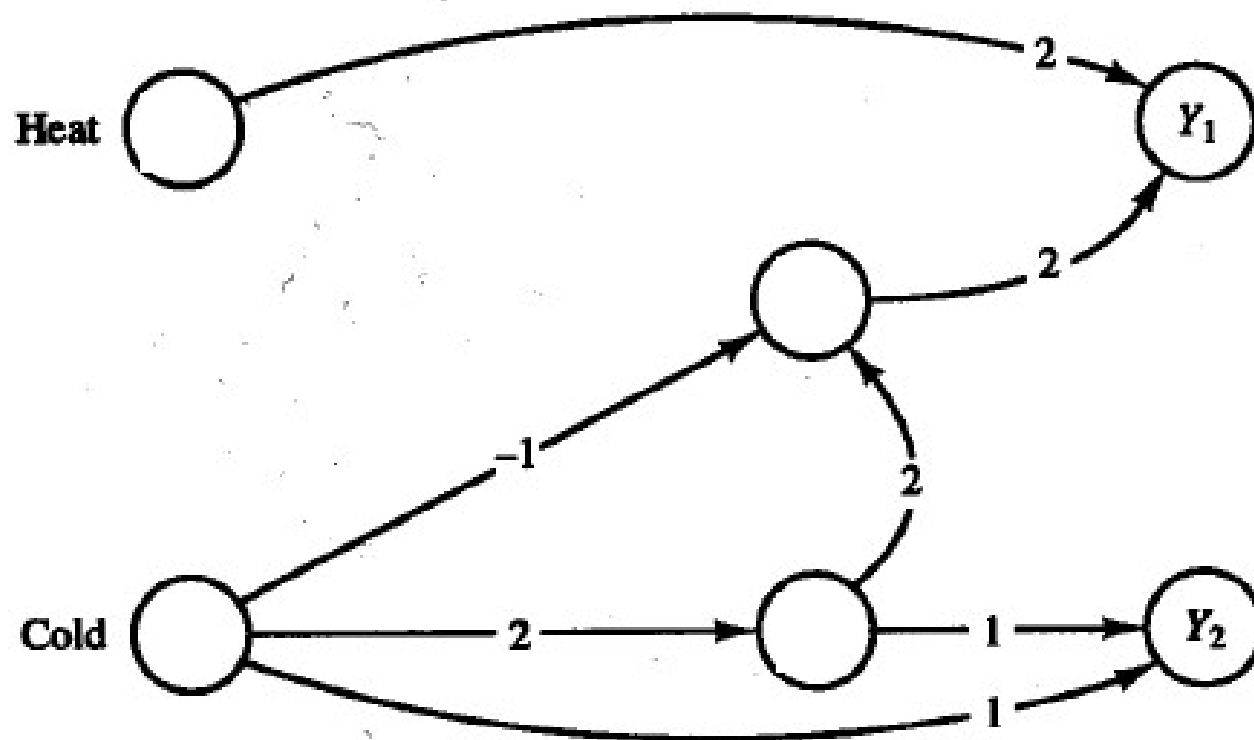
t=3



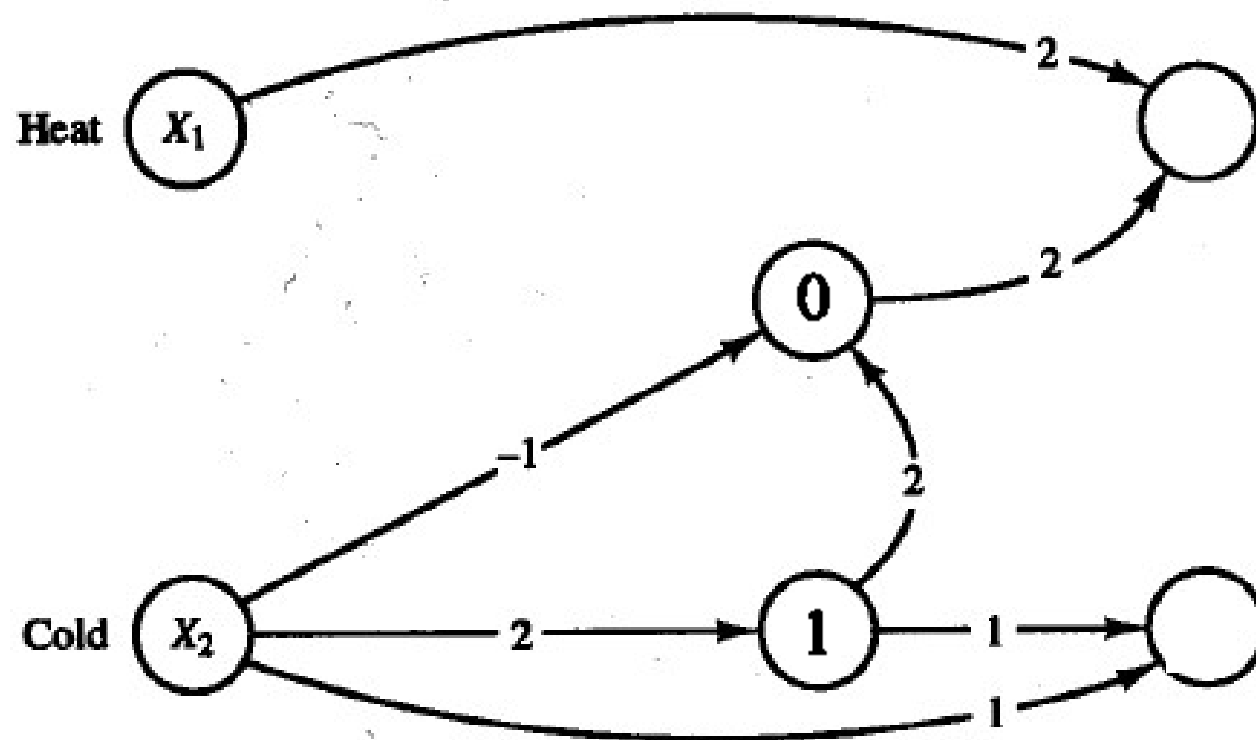
Cold Stimulus (two step)



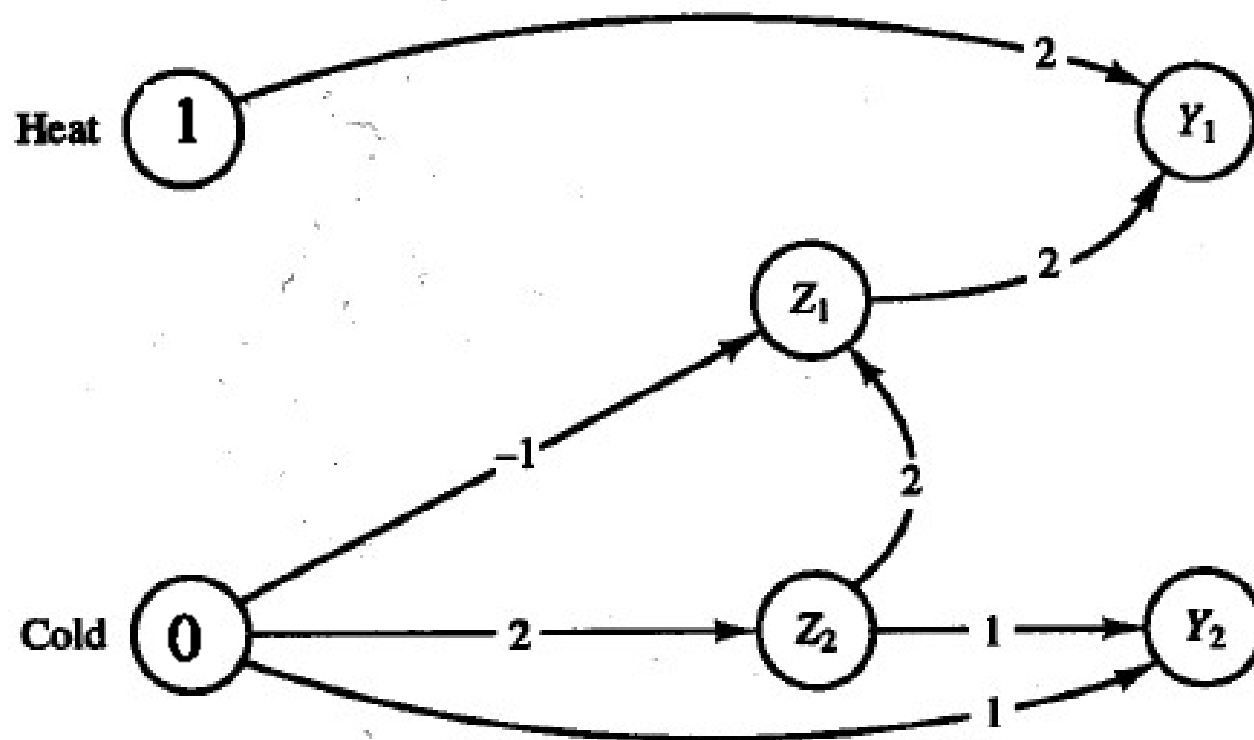
$t=1$



$t=2$



Hot Stimulus (one step)



$t=1$

